Software Requirements Specification

Online Bookstore

An e-commerce platform for buying and selling books online

1. Introduction

This document specifies the Functional Requirements (FR) and Non-Functional Requirements (NFR) for the Online Bookstore system, an e-commerce platform that allows buyers to browse and purchase books, sellers to list and manage inventory, and admins to moderate the platform. Requirements are derived from the User, System, and Domain requirements identified during requirements elicitation.

2. Functional Requirements

The table below lists the functional requirements that define the specific behaviours and features the system must support.

Req ID	Functional Requirement Description
FR-1	Allow users to register and log in using email/phone with password authentication.
FR-2	Allow buyers to search books by title, author, ISBN, genre, and price range.
FR-3	Display book details including price, availability, seller, condition, and reviews.
FR-4	Allow buyers to add/remove items from a shopping cart.
FR-5	Allow buyers to place an order and complete payment through an integrated payment gateway.
FR-6	Send a confirmation email/notification whenever an order is placed.
FR-7	Allow sellers to add, edit, and remove book listings.
FR-8	Allow sellers to view and manage incoming orders for their listings.
FR-9	Allow buyers to track order status (placed, shipped, out for delivery, delivered).
FR-10	Allow buyers to rate and review purchased books.
FR-11	Allow buyers to apply discount codes/coupons at checkout.
FR-12	Allow admins to approve, suspend, or remove listings and user accounts.
FR-13	Generate an order invoice/receipt after successful payment.
FR-14	Allow users to reset a forgotten password via email/OTP verification.
FR-15	Maintain a wishlist feature allowing buyers to save books for later.

3. Non-Functional Requirements

Non-functional requirements are grouped into Product, Organisational, and External categories as per the requirements specification.

3.1 Product Requirements

Req ID	Category	Non-Functional Requirement Description
NFR-1	Performance	Page load time shall not exceed 2 seconds under normal load (up to 500 concurrent users).

Req ID	Category	Non-Functional Requirement Description
NFR-2	Reliability	The system shall maintain 99.5% uptime, measured monthly.
NFR-3	Scalability	Support horizontal scaling to accommodate seasonal spikes without redesign.
NFR-4	Usability	A first-time user shall complete a purchase within 5 minutes without external help.
NFR-5	Portability	Accessible via modern browsers (Chrome, Firefox, Safari, Edge) and responsive on mobile.
NFR-6	Security	All sensitive data (passwords, payment info) shall be encrypted in transit (TLS 1.2+) and at rest.

3.2 Organisational Requirements

Req ID	Category	Non-Functional Requirement Description
NFR-7	Process Standards	Development shall follow Agile/Scrum with 2-week sprints and code review before merge.
NFR-8	Implementation	Backend shall use the team's chosen framework; database shall use SQL for transactional integrity.
NFR-9	Maintainability	Codebase shall maintain a minimum of 70% unit test coverage.

3.3 External Requirements

Req ID	Category	Non-Functional Requirement Description
NFR-10	Legal/Regulatory	The system shall comply with data protection laws applicable to the regions it serves.
NFR-11	Interoperability	Payment and shipping integrations shall use standard, documented third-party APIs.
NFR-12	Availability	Customer support channels (chat/email) shall respond within 24 hours.